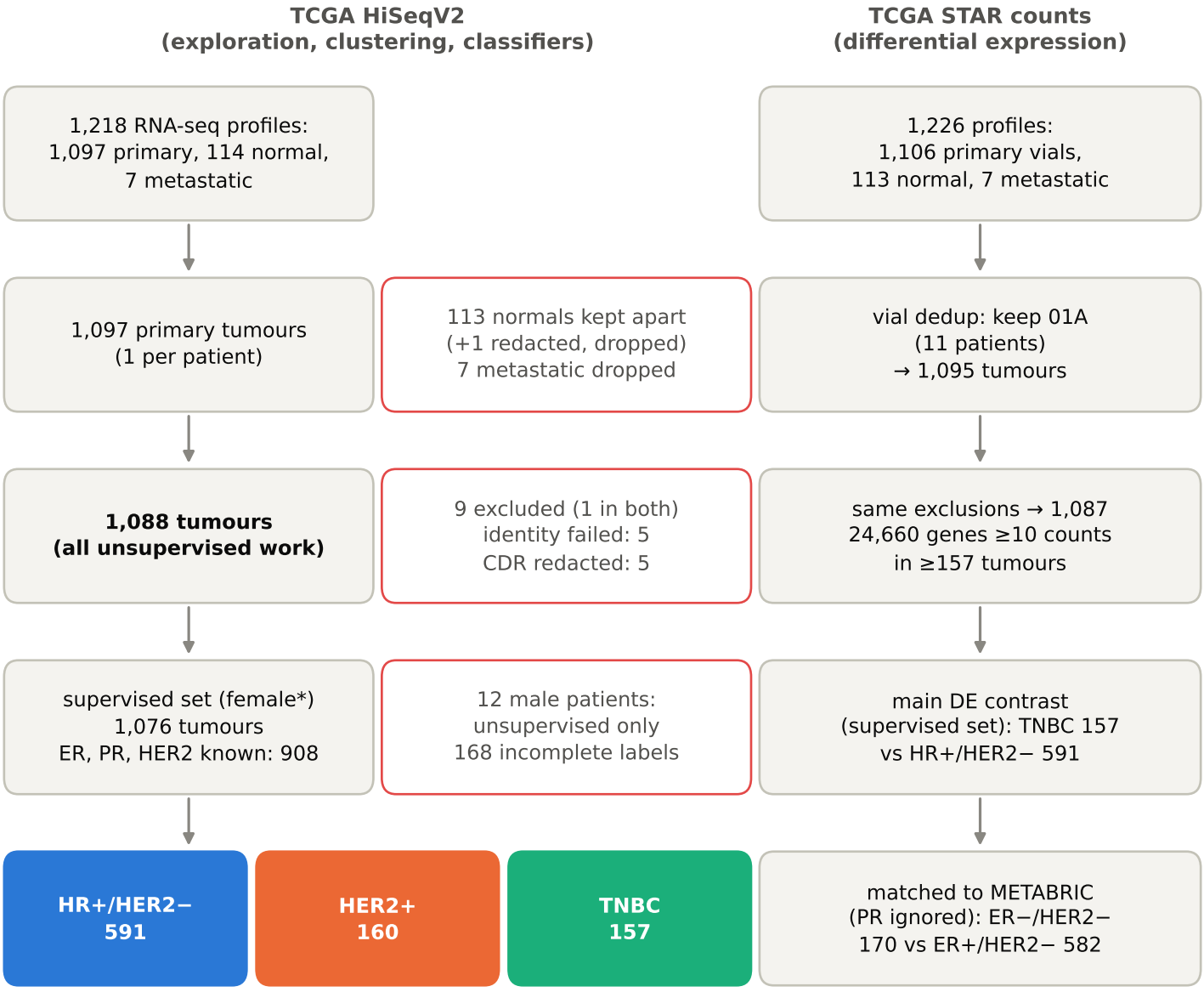
Sample flow: from downloaded files to analysis groups



Validation cohorts (boxes on slide 3). TCGA RPPA: 730 supervised tumours with RNA and protein (TNBC 109, HER2+ 114, HR+/HER2− 395); 91 total-protein antibodies mapped to genes. METABRIC: 1,980 tumours (Illumina HT-12, 18,936 genes shared with TCGA), labels from other assays only (ER by IHC, HER2 by SNP6 copy number): ER+/HER2 not gained 1,220, ER−/HER2 not gained 286, HER2 gained 426.

HER2 rule: FISH first; otherwise IHC score (3+ positive, 0/1+ negative, 2+ without FISH = missing, IHC 3+ with negative FISH = missing); IHC status text only when no score; status/score contradictions = missing. Groups need ER, PR and HER2 known.

Identity check: a tumour's HiSeqV2 and STAR profiles must correlate at $\rho \geq 0.80$, above the 99th percentile of between-patient ρ (0.87), and be each other's best match. *Recorded sex, or XIST/RPS4Y1 expression when blank (n=2). METABRIC's ER−/HER2-not-gained group is not called TNBC (no IHC PR).